Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	**Semester**	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	**Batch**	2023–2027

Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

Objective

- To implement a **Decision Tree classifier**.
- To extend the Decision Tree into a **Random Forest ensemble model**.
- To study the impact of hyperparameters on overfitting and generalization.
- To **select optimal hyperparameters using 5-Fold Cross-Validation**.
- To compare single-tree and ensemble-tree models.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: https://archive.ics.uci.edu

Theory

Decision Tree Classifier

A Decision Tree is a supervised learning model that recursively splits the feature space using impurity measures to form decision rules.

Key Concepts:

- Gini Index and Entropy
- Tree depth and node splitting
- Overfitting in deep trees

Limitation: Decision Trees are prone to high variance and overfitting.

Random Forest Classifier

Random Forest is an ensemble learning technique that combines multiple decision trees trained on bootstrapped samples.

Key Ideas:

- Bootstrap aggregation (bagging)
- Random feature selection
- Majority voting

Advantage: Reduces variance and improves generalization compared to a single Decision Tree.

Steps for Implementation

1. Load the dataset and encode class labels.
2. Perform Exploratory Data Analysis:
   - Class distribution
   - Feature correlation analysis
3. Split the dataset into training and testing sets (80–20).
4. Train a Decision Tree classifier.
5. Define a hyperparameter search space.
6. Perform **5-Fold Cross-Validation** to evaluate each hyperparameter combination.
7. Select the hyperparameters that yield the best average cross-validation performance.
8. Train a Random Forest classifier.
9. Repeat cross-validation-based hyperparameter selection.
10. Compare both models using evaluation metrics.

Hyperparameters to be Explored

Decision Tree

- `criterion`
- `max_depth`
- `min_samples_split`
- `min_samples_leaf`

Random Forest

- `n_estimators`
- `max_depth`
- `max_features`
- `bootstrap`

Hyperparameter Tuning Results

Decision Tree Cross-Fold Results

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
Gini	5	93	0.93
Entropy	5	94	0.94

Random Forest Cross-Fold Results

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	5	sqrt	96	0.96
100	5	sqrt	97	0.97

5-Fold Cross-Validation Performance Comparison

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.92	0.93	0.94	0.91	0.93	0.93
Random Forest	0.96	0.97	0.98	0.97	0.96	0.97

Evaluation Metrics

- Accuracy
- Precision
- Recall

- F1-score
- Confusion Matrix
- ROC Curve and AUC

Observation Questions

- How does tree depth affect overfitting in Decision Trees?
- Which hyperparameter had the greatest impact on performance?
- How does Random Forest improve generalization?
- Did ensemble learning always improve performance? Why or why not?

Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation. Hyperparameters were selected based on average cross-validation performance, ensuring robust generalization. The results demonstrate that Random Forest reduces variance and improves stability compared to a single Decision Tree.

References

- Scikit-learn: Decision Trees
- Scikit-learn: Random Forest
- UCI Dataset

Conclusion

Decision Tree and Random Forest models were implemented using the breast cancer dataset. 5-fold cross-validation was used to select optimal hyperparameters.

The Decision Tree achieved an average accuracy of around 93%, while the Random Forest achieved approximately 97%. Random Forest performed better due to ensemble learning, reducing overfitting and improving generalization.

Thus, Random Forest is more robust and accurate compared to a single Decision Tree.